

The reality stack

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A stack is layers where each one rests on the one below — like a tech stack in computing, rock strata in geology or floors in a building.

Why define a stack? The stack exists to turn “is this just the way things are?” into “which layer does this object sit at, and therefore how contingent, how mutable, and how accountable is it?”

Why a reality stack and not *the* reality stack? Because “the” would imply there is one correct way to slice reality, there isn’t. This is just one lens.

A note on motivation. The framework is descriptive in form but normatively motivated: it exists to make a particular claim about inequality arguable. The reader should hold both hands visible.

This is not original metaphysics — it is a teaching device for readers who need the distinctions made explicit before the political argument lands.

Layers of reality

There are three layers and one cross-section.

The **physical layer** is invariant law. Gravity, thermodynamics, the speed of light. You cannot negotiate with it. It gives rise to everything above but does not determine it. Examples: gravity, the speed of light, the fact that water freezes at 0°C.

The **material layer** is the physical world in the mode of human encounter: bodies, biology, climate, geography, and the infrastructure we have built into it. It is not philosophical materialism (the doctrine that everything reduces to physics) — it is physics considered through the lens of what humans need, what they are stuck with, and what they have made. Examples: the calorie requirement of a human body, the climate where you were born, the fact that there is a river here and not there, the concrete your city is made of.

The **social layer** is the constructed layer — arrangements humans make collectively, constrained by the material but not determined by it. Two societies with identical material conditions can organise themselves in opposite ways, and historically have, sometimes apparently on purpose. It is where power, roles, norms, and institutions live — and it is the layer people tend to mistake for one of the layers below. Examples: which side of the road you drive on, marriage, queuing, money.

The **agent** is not a fourth layer. An agent is a cross-section — a single point where all three layers meet. An agent has a physical body, material needs, and a social layer carried inside as language, shame, politeness. What is purely agent? Very little — a flinch, a heartbeat, a refusal. And yet the cross-section is the only site from which any of it can be questioned. Agents, acting together, reshape the social layer, which reshapes the material, which reshapes what is possible.

Locating a thing in the stack

To test where a thing sits, ask: *if the other layers were varied independently, would this thing vary with them, or hold constant?* Gravity holds across all societies and all bodies — physical. The calorie requirement of a human body holds across all societies but varies across species — material. Which side of the road cars drive on varies across societies within identical material conditions — social.

The test is not a decision procedure. Many interesting cases — gender, race, disease, addiction — pass it ambiguously, and that is the point: the framework surfaces the dispute rather than resolves it. Where the layer-assignment is contested, the framework names the contest. That is less than a theory and more than a vibe.

Natural and real

We are suspicious of the word **natural** because it often does covert work — moving a social claim down to the physical layer, where it can no longer be argued with. “Hierarchy is natural” makes hierarchy a law of gravity. “It is human nature to be selfish” puts a collective behaviour on the physical layer despite massive anthropological evidence to the contrary. The word has legitimate uses — naturalistic ethics, virtue theory, evolutionary biology all use it without sleight of hand. The suspicion is of the rhetorical move, not the word.

We are careful with the word **real**. Everything in the stack is real in some sense, but not in the same sense. A desk is real in a way that does not depend on anyone believing in it. Money is real in a way that entirely depends on collective uptake — if everyone stopped treating the green paper as money overnight, it would stop being money. Call these *brute facts* and *institutional facts*. The distinction matters: institutional facts are real but ontologically dependent on continued collective practice. “Money is real” is true. “Money is real in the same way a desk is” is false. The question “is it real?” is rarely interesting. The question “on what does its reality depend?” almost always is.

Boundaries between layers

The boundaries are fuzzy, and the fuzziness is load-bearing. The framework cannot settle any dispute whose resolution turns on where a thing sits — it can only make the placement visible.

The **physical/material boundary** is where physics becomes relevant to human life. A hurricane is physical; it becomes material when it threatens a city’s food and water. The **material/social boundary** is where biology becomes arrangement. A river is material; whether it is a border or a highway is social. Upper body strength is material; the social role built on top of it is not. The **social/agent interface** is the subtlest. The social is constructed by agents and absorbed by agents. The interface is where human arrangements start to feel like nature, where they become part of who you are rather than something you do.

Three axes, not one

Three spectrums run through the stack. They are related but not interchangeable.

Contingency — whether something could have been otherwise. Rerun human history from similar starting conditions; what comes out the same is necessary, what comes out differently is contingent. The physical layer has zero contingency. The material has some. The social has a great deal.

Mutability — how readily something can be changed. You cannot change gravity. Material conditions take generations — clear a forest, build a dam, industrialise. Social arrangements can change in a lifetime. An agent’s position can change in a moment.

Accountability — who bears responsibility. This is the axis the other two do not straightforwardly produce. A thing can be contingent and mutable and yet have no locatable individual at fault. Historical slavery was contingent in principle and mutable in principle, but moral responsibility under ideology, under incomplete information, under collective-action constraints, is not captured by “someone chose this.” Responsibility can be collective and forward-looking

even when no individual chose the outcome. The stack's third axis is not "who caused it" but "who, going forward, is situated to change it."

Contingency is the descriptive axis, mutability is the practical axis, accountability is the normative axis. They can come apart. A thing can be contingent but practically frozen — path-dependent lock-in. A thing can be mutable in principle but collectively unaccountable — no agent is situated to change it alone. The interesting political arguments are always about which axis a dispute is on.

How the layers interact

The layers do not sit passively on top of each other. The interesting stuff — most of politics, most of history — happens at the interfaces.

Upward constraints. The physical constrains the material — you cannot build a skyscraper in the middle of the ocean. The material constrains the social — you cannot have a farming society in a desert. The social constrains the agent — you are born with the capacity to learn any language on earth, but which one you actually speak is entirely social.

Downward causation. The upper layers reach down and reshape the lower ones. You plant a garden — agent reshapes material. A village decides to build a well — social reshapes material. A revolution overthrows a government — agents reshape the social. The stack is not one-directional. This is what makes it a stack and not just a hierarchy.

Downward causation is philosophically contested. If the physical is causally closed, non-physical causes either overdetermine outcomes or are epiphenomenal — they do no work that physics is not already doing. The stack assumes *non-reductive physicalism*: higher-level facts have causal purchase even though they supervene on physical ones. This is the working assumption of most social science. It is adopted here rather than defended. A reader who rejects it will find the rest of the apparatus unmotivated.

Naturalisation. Something from the social layer is smuggled down to the physical. "Boys will be boys" — social behaviour relocated to the physical in four words. "It's human nature to be selfish" — the most popular naturalisation of all, despite massive anthropological evidence against it. It is a move to escape accountability by crossing into a layer where accountability does not apply.

Material lock-in. The social builds material infrastructure that then constrains future social choices. Once you have built a car-dependent city, the material layer locks in a social arrangement. A railway gauge, once laid, forces everything that runs on the tracks to match. The social created it; the material now maintains it. This is why it is much easier to imagine a different society than to build one: the old one has been poured into concrete.

Internalisation. The social becomes the agent when a person absorbs a construction as part of who they are. You lower your voice in a library — nobody is enforcing it, but you would feel like a criminal if you didn't. You feel guilty on a day off — the boss is gone but the feeling stayed. Language may be the deepest case. You did not choose your first language. It arrived before you could consent to it, it structured how you think, and you cannot unlearn it. Entirely social, yet permanent. Internalisation is not total: a trans person who rejects the gender assigned to them is demonstrating that the cross-section is not a passive receiver.

Emergence. Agent-level actions produce patterns no agent intended. A path forms across a field. Billions of decisions to burn coal produce a warmer planet. Indoor plumbing starts as a

luxury and becomes a need. The social layer emerges from agents but is not reducible to them — which is why individual goodwill is not enough to fix collective outcomes.

Inter-layer feedback. An material difference (upper body strength) becomes a social category (gender roles), which is internalised by agents (gender identity), who then perform it in ways that reinforce the social category, which gets mistaken for a physical fact. The signal bounces up and down the stack, amplifying at each pass. By the time it has made a few trips, nobody remembers that it started as a small material difference rather than an immutable law of nature.

A stress test: gender

The easy examples make the framework look cleaner than it is. Here is a hard case.

Is gender physical, material, or social? The honest answer is that the framework does not decide — it names where the disagreement is.

A **biological essentialist** places gender at the material layer: sexual dimorphism, hormonal differences, reproductive asymmetry. If gender is material, much of what follows — gendered division of labour, certain forms of hierarchy — is constrained rather than chosen.

A **strong social constructionist** places gender at the social layer: a system of roles, expectations, and performances with no uniform biological signature. If gender is social, everything built on it is open to contest.

A **feedback view** — which is what this essay leans toward — takes gender as starting at the material layer with a small, diffuse difference, and escalating through repeated passes between the material, social, and agent layers until the social category towers far above any material signal that could motivate it. On this view the earlier two positions are not so much wrong as incomplete: each names one layer of a structure whose whole behaviour only becomes visible in the loop.

The framework's value here is not that it adjudicates. It is that it forces the disputants to say which layer they are placing the thing at, and to defend that placement. That is more than most arguments about gender offer.

Two examples

Walking. *Physical:* bipedalism, gravity, friction. *Material:* terrain — mud, mountain, flat plain, river crossing. *Social:* sidewalks, which side of the pavement you walk on, where women can walk alone at night, borders, trespass law. *Agent:* where you choose to go, whether you jaywalk, whether you cross a border illegally. A border is invisible at the physical and material layers — the ground is the same on both sides — but at the social layer it is one of the most powerful structures on earth. People die crossing borders. The dirt does not know.

Oil. *Physical:* chemistry — heat and pressure convert buried organisms into hydrocarbons. *Material:* the oil sits in specific places — the Persian Gulf, the North Sea, West Texas — unevenly distributed by geological accident. *Social:* who owns it, who profits, the decision to build a civilisation around the internal combustion engine. The same barrel could power a public bus or a private jet — that is a social choice. *Agent:* you drive, fly, cycle. Here the layers fold: agent-level consumption, aggregated, reshapes the material — CO₂ accumulates, the climate warms — which constrains future social choices. The stack operating as a loop.

Objections

A framework this ambitious attracts specific objections. The serious ones:

The stack is a pedagogical metaphor, not structure. Reality is closer to a tangled hierarchy than to ordered dependence; every “downward causation” sentence already admits this. The defence is modest: the stack is a simplification that keeps the explanation tractable. A reader who wants the full graph is right to want it; this essay is not providing it.

Layer assignment does the argumentative work, and is not independently defended. Charges of naturalisation presuppose the thing is social. This is fair. The response is the layer-membership test and the stress test on gender: the framework diagnoses rather than decides. Where the diagnosis and the defence of placement both fail, the framework has nothing to say.

Downward causation is assumed, not earned. Acknowledged. The exclusion problem — that if physics is causally closed, higher-level causes do no independent work — is named, not engaged. A full defence of non-reductive physicalism is out of scope.

The earlier treatment of “real” was too fast. The brute/institutional distinction above replaces it. Institutional facts are real but ontologically dependent; that is the substantive claim “everything is real” was hiding.

Accountability was being smuggled in. The move is no longer “contingent, therefore accountable.” It is “contingent, therefore open to forward-looking collective responsibility.” Weaker, and more defensible.

Normativity masqueraded as description. The framework is descriptive in form and normatively motivated. This is now stated up front. A reader who rejects the motivation can reject the framework on that basis.

Science thins as you move up

Science’s power decreases as you move up the stack — each layer adds contingency, and science is best at the necessary. The physical layer is the most understood and still deeply incomplete. The material layer is well-covered by biology and earth science but cannot predict tipping points. The social layer is where science thins dramatically — economics claims to be a science but its predictions are poor, sociology is descriptive more than predictive, and there is no reliable theory of why some societies become egalitarian and others hierarchical. At the social and agent levels, where this argument lives, science offers tools but not answers. The rest has to come from somewhere else.

So when you catch yourself asking “is this just the way things are?” — locate the thing in the stack. If it sits at the physical layer, accept it. If it sits anywhere above, the question has an answer: it is this way because of the layer it sits at, the path it took, and the feedback that locked it in. None of that is destiny. That is the whole point of naming the layers.